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General Comment

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Attachments

Pindrop Response to RFI on Development of an Artificial Intelligence (FINAL 03-14-25)



1115 Howell Mill Road, Suite 700
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March 14, 2025

**Re: Pindrop Security, Inc. (Pindrop) Response to
RFI on Development of an Artificial Intelligence (AI) Action Plan**

“With the right Government policies, we can solidify our position as the global leader in AI and secure a brighter future for all Americans.”

- President Donald J. Trump, Executive Order 14179, January 23, 2025

Pindrop appreciates the opportunity to respond to the Trump Administration’s request for information on the development of an AI Action Plan. Pindrop applauds First Lady Melania Trump for recently highlighting AI deepfakes at the State of the Union when she brought a 15-year-old victim of them, spotlighting both the perils of deepfakes and the opportunities to combat them.

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1. Background on Pindrop

With shared concern and passion, Pindrop deploys AI solutions to help businesses and individuals defend themselves against increasingly sophisticated AI-oriented threats, like deepfakes.

Pindrop is a privately held U.S.-based company, founded in 2011 and headquartered in Atlanta, GA.

Pindrop solutions are trusted by some of the largest U.S. companies, including the country’s major banks, retailers and largest insurers. Our voice-authentication and fraud detection technology analyzes over 1.2 billion calls per year.

Pindrop technology provides real-time detection solutions – bolstered by one of the industry's most extensive consortiums of known fraud actors, with more than 1.3 million confirmed fraudster phone numbers.

Pindrop's deepfake detection technology, *Pindrop Pulse*, is the culmination of more than a decade of research in voice security.¹ A leader in liveness detection, we have set the standard with 99% accuracy in determining *real* voices versus *fake* voices.² Pindrop also possesses expertise in video as well as voice security, harnessing advanced Artificial Intelligence (AI) and Machine Learning (ML).

Pindrop Pulse was developed using over 25 patented and patent-pending innovations in deepfake technology and has received numerous recognitions over the years, including:

- Pindrop was named as the large organization winner of the FTC 2024 Voice Cloning challenge.³
- NPR conducted a blind test in 2024 of various deepfake detection systems and *Pindrop Pulse* was recognized as having the highest accuracy. Pindrop's accuracy was reported as 40 percentage points more effective than the nearest competitor by NPR.⁴
- Pindrop researchers have been regular participants in the NIST Speaker Recognition Evaluation (SRE)⁵ since 2012⁶ as well as the annual Automatic

¹ Tianxiang Chen, Avrosh Kumar, Parav Nagarsheth, Ganesh Sivaraman, and Elie Khoury. "Generalization of Audio Deepfake Detection." In *Odyssey*, pp. 132-137. 2020. Elie Khoury, Tomi Kinnunen, Aleksandr Sizov, Zhizheng Wu, and Sébastien Marcel. "Introducing i-vectors for joint anti-spoofing and speaker verification." *Interspeech*. 2014. Tianxiang Chen, Elie Khoury, Kedar Phatak, and Ganesh Sivaraman. "Pindrop labs' submission to the ASVspoof 2021 challenge." *Proc. 2021 Edition of the Automatic Speaker Verification and Spoofing Countermeasures Challenge (2021)*: 89-93. Tianxiang Chen, and Elie Khoury. "Spoofprint: a new paradigm for spoofing attacks detection." In *2021 IEEE Spoken Language Technology Workshop (SLT)*, pp. 538-543. IEEE, 2021.

² <https://www.pindrop.com/article/effective-signal-modified-deepfakes-liveness-detection/>

<https://www.pindrop.com/article/truth-about-zero-day-deepfake-attacks/>

³ <https://www.ftc.gov/news-events/contests/ftc-voice-cloning-challenge>

⁴ <https://www.npr.org/2024/04/05/1241446778/deepfake-audio-detection>

⁵ <https://www.nist.gov/itl/iad/mig/speaker-recognition>

⁶ Rahim Saeidi, Kong-Aik Lee, Tomi Kinnunen, Tawfik Hasan, Benoit Fauve, P-M. Bousquet, Elie Khoury et al. "I4U submission to NIST SRE 2012: A large-scale collaborative effort for noise-robust speaker verification." in *Proc. Interspeech*. 2013.

Speaker Verification Spoofing and Countermeasures (ASVspoof) challenge⁷ series since 2017.⁸

- Our research on the topic of audio deepfake source tracing has also progressed, and a paper with our findings was accepted at the Interspeech 2024 conference.⁹

In developing this technology, Pindrop has created a proprietary database of 27M+ audio utterances from 500+ generative artificial intelligence (genAI) systems.

Pindrop was founded by Dr. Vijay Balasubramaniyan, Dr. Paul Judge and Dr. Mustaque Ahamad, and is venture-backed by Andreessen Horowitz, Citi Ventures, Felicis Ventures, CapitalG, GV, IVP and Vitruvian Partners. Pindrop is proud of its strong American tech workforce, with over 90% of Pindrop's workforce in the United States.

2. Threats to the U.S. from AI-Generated Deepfakes

This RFI appropriately asks for recommended “concrete AI policy actions needed to address the topics raised” in the RFI. In order to provide such concrete recommendations, it is important to understand the risks posed with the misuse of AI, and the threats that are growing exponentially.

Due to rapid advances in technology, it now takes only 2-5 seconds of a person's voice to create a credible clone of that voice. While there was only one Text to Speech (TTS) engine in 2018, today there are more than 500 TTS engines in 2025 (with some from China). This increase in accessibility and sophistication allows rapid scalability.

American institutions, businesses, and citizens are vulnerable to attacks by bad actors using AI-generated deepfakes to conduct financial scams, cybersecurity hacks and intellectual property thefts. Easily scalable and low-cost AI tools have opened doors for both good and bad applications. With the surge in AI and deepfakes, we believe that good AI such as deepfake detection tools are needed to combat bad AI, ensuring that American enterprises reap the full potential of AI, mitigate harm to their organizations and customers, and maintain America's AI leadership globally.

⁷ <https://www.asvspoof.org/>

⁸ Tianxiang Chen, Avrosh Kumar, Parav Nagarsheth, Ganesh Sivaraman, and Elie Khoury. "Generalization of Audio Deepfake Detection." In *Odyssey*, pp. 132-137. 2020.

⁹ <https://www.pindrop.com/article/research-audio-deepfake-source-tracing/>

Financial Services

The recommendations that we set forth below in Section 3 are informed by the experience that we have gained over the years of developing technology to counter deepfakes and attacks especially on financial institutions.

According to Deloitte's Center for Financial Services, genAI could enable fraud losses in the U.S. to reach \$40B by 2027 (up from \$12B in 2023).¹⁰ Pindrop's internal data reflects that the number of deepfake calls in its customers' call centers grew from 0.12% of all calls in Q1 2024 to nearly 1% by the end of 2024, a 683% growth during the year.

By late 2023, with AI tools (e.g., ChatGPT) widely available, synthetic voices began to sound more “human”. By 2024, Pindrop internal data suggests there are seven deepfake attacks per day (up from one attack per month in 2023).

This shift represents a fundamental change in how we approach fraud and digital security now and in the future: the question “are you the right human?” is no longer sufficient. It needs to be augmented with “are you even a real human?” Of U.S. companies surveyed by Deloitte, nearly 26% reported a deepfake incident targeting financial and accounting data in 2024.¹¹ Of those companies surveyed, 51.6% expected the number and size of attacks to increase in the next 12 months.¹² Looking ahead, the fundamental question at the center of every business interaction will be, “is this a machine, is this a potential fraudster, or is this my genuine customer?”

An AI Action Plan should take into consideration the current threat environment that many companies are facing, especially the financial services sector. Today, Pindrop is helping customers in the financial services sector combat four kinds of attacks by bad actors who use gen AI to produce deepfakes over phone calls and video interactions:

- 1) Individual Consumer Account Takeover: Currently, these attacks happen predominantly in call centers via the phone channel. Pindrop solutions have detected and helped prevent instances of bad actors using synthetic voices to impersonate genuine consumers in U.S. financial institutions and

¹⁰ <https://www2.deloitte.com/us/en/insights/industry/financial-services/financial-services-industry-predictions/2024/deepfake-banking-fraud-risk-on-the-rise.html>

¹¹ <https://www2.deloitte.com/content/dam/Deloitte/us/Documents/Advisory/us-generative-ai-and-the-fight-for-trust.pdf>

¹² <https://www2.deloitte.com/content/dam/Deloitte/us/Documents/Advisory/us-generative-ai-and-the-fight-for-trust.pdf>

retailers. U.S. media have reported schemes targeted at senior citizens, often called Phantom Hacker scams.¹³

- 2) *Fraudulent Corporate Money Transfers:* Multiple instances of fraud and attempted fraud have been reported in the press involving accidental transfers of large sums of money at well-known international corporations. Early in 2024, an employee of UK engineering firm Arup made a seemingly routine transfer of millions of company dollars following a video call with senior management.¹⁴ Except, it turned out, the employee had not been talking to Arup managers at all but to deepfakes created by artificial intelligence. The employee had been tricked into sending \$25 million to criminals. Similar attempts have been reported at WPP¹⁵ and Ferrari.¹⁶
- 3) *Remote Interview Impersonation:* According to an article in the *Wall Street Journal*, more than 300 U.S. companies unknowingly hired foreign nationals with ties to North Korea for remote IT work.¹⁷ Several cyber security companies, such as Knowbe4¹⁸ and Vidoc Security Labs¹⁹ have reported attempts by foreign nation-states to position personnel for hiring using fake identities and AI-generated deepfakes. Last month, Pindrop's recruiting team identified sophisticated attempts by fraudsters to impersonate a job candidate during a virtual interview for one of our remote engineering roles. Our analysis uncovered that the individual was utilizing real-time face-swapping technology powered by gen AI, effectively masking their true identity with an entirely different, unknown face in multiple interviews.
- 4) *Infiltration of Corporate Meetings:* Several of Pindrop's customers have expressed concerns about being vulnerable to espionage attempts by bad actors using deepfakes to join internal video meetings (e.g., company town halls, review of financial information, planning for sensitive company

¹³ <https://www.foxnews.com/tech/dont-let-ai-phantom-hackers-drain-your-bank-account>

¹⁴ <https://www.weforum.org/stories/2025/02/deepfake-ai-cybercrime-arup/>

¹⁵ <https://nypost.com/2024/05/10/business/ceo-of-wpp-falls-victim-to-deepfake-scam/>

¹⁶ <https://sloanreview.mit.edu/article/how-ferrari-hit-the-brakes-on-a-deepfake-ceo/>

¹⁷ <https://www.wsj.com/articles/deepfakes-fraudsters-and-hackers-are-coming-for-cybersecurity-jobs-e2a76d06>

¹⁸ <https://thecyberexpress.com/knowbe4-fake-employee-north-korean-hacker/>

¹⁹

https://www.theregister.com/2025/02/11/it_worker_scam/?utm_medium=share&utm_content=article&utm_source=linkedin

transactions) or meetings with external partners, especially those with presence in international countries. There are more than 1 billion daily active users across popular virtual conferencing platforms, making video meetings the primary mode of business interaction.

National Security

Deepfakes have the potential to compromise national security. Bad actors are using easily available technology to create deepfakes of public figures to spread disinformation, weaken our security posture, and undermine trust in elections/democracy. For example:

- In 2024, the Chairman of the U.S. Senate Foreign Relations Committee, Ben Cardin, was a target of a deepfake over a Zoom call when a figure appeared on the call and impersonated Dmytro Kuleba, the former Foreign Minister of Ukraine.²⁰ Kuleba demanded an opinion on sensitive foreign policy questions such as whether the Senator supported firing long-range missiles into Russian territory.
- Another example, as noted earlier, is that adversaries such as North Korea are using AI-generated deepfakes to surreptitiously place workers into IT roles at American companies.²¹ In a December, 2023 Cybersecurity Information Sheet, the NSA provided that “Threats from synthetic media, such as deepfakes, present a growing challenge for all users of modern technology and communications, including National Security Systems (NSS), the Department of Defense (DOD), the Defense Industrial Base (DIB), and national critical infrastructure owners and operators.”²²

3. Pindrop Policy Recommendations

Pindrop agrees that “bad AI” can be used in many harmful ways, and we welcome the White House’s leadership in finding ways to use AI to address these challenges. We recommend that the Trump Administration’s AI Action Plan account for and include the following principles and priorities.

²⁰ <https://www.nbcnews.com/politics/congress/ben-cardin-targeted-apparent-deep-fake-call-dmytro-kuleba-rcna172776>

²¹ <https://www.cio.com/article/3542825/is-your-coworker-a-north-korean-hacker-how-ai-impersonation-is-compromising-the-workforce.html>

²² <https://media.defense.gov/2023/Sep/12/2003298925/-1/-1/0/CSI-DEEPFAKE-THREATS.PDF>

Advance real-time deepfake detection solutions to counter threats to American government, businesses and consumers

To address the growing threat of AI-enabled deepfakes and other audio-based attacks by foreign adversaries, Pindrop suggests:

- White House and government agencies should adopt real-time audio deepfake detection technologies in relation to critical infrastructure. These advanced detection solutions, which flag synthetic or manipulated audio in real-time, are critical for safeguarding communication networks, preventing disinformation campaigns, and mitigating security risks. Such an approach would align with other recent federal efforts to secure AI ecosystems in relation to national security. By prioritizing these technologies, the U.S. can protect its digital infrastructure from foreign exploitation while fostering innovation in securing AI applications.
- Industries critical to American infrastructure, such as financial services, defense, utilities, and cyber security should use deepfake detection technology to prevent bad actors, including nation-states, from penetrating critical infrastructure and industries. Deepfake detection should be used for all high-risk virtual meetings with external participants, including recruiting, international suppliers, and partners.
- Mobile operating systems and device manufacturers should offer ‘on-device’ deepfake detection to protect U.S. citizens from bad actors carrying out misinformation or financial scam campaigns.

Strengthen Know Your Customer protocols

To strengthen identity verification processes and combat identity fraud, Pindrop advocates for the integration of liveness detection technologies into Know Your Customer (KYC) protocols across industries like banking, fintech, and telecommunications.

- By leveraging advanced machine learning and computer vision techniques, this technology can detect spoofing attempts, deepfakes, and other presentation attacks with high accuracy.
- Federal guidelines from agencies like the FCC and NIST could establish standards for both active and passive liveness detection methods, ensuring compliance with Anti-Money Laundering (AML) laws and enhancing customer due diligence. This approach would not only mitigate risks from identity theft and fraud but also protect against potential exploitation by foreign actors, ensuring the integrity of critical financial systems while fostering trust in digital transactions.

Promote “liveness detection” as a technical standard

Currently there is no unified standard regarding voice authentication. The White House and NIST should consider:

- Adopting "liveness detection," which makes a determination of whether the voice is human or machine, as a standard for voice authentication by federal regulators such as the FCC, NIST, and others. Doing so would significantly enhance the security and reliability of telecommunications and financial services.
- By analyzing features like natural variability in speech, spectral characteristics, and temporal patterns, liveness detection can differentiate between live voices and synthetic attempts with high accuracy. Incorporating liveness detection into regulatory frameworks would further efforts to mitigate fraud in voice networks, complementing existing measures like STIR/SHAKEN for caller ID authentication.
- Standardizing liveness detection would also promote trust in voice-based systems while fostering innovation in secure biometric authentication. As bad actors increasingly exploit advanced technologies to bypass traditional safeguards, liveness detection offers a proactive defense mechanism. Federal regulators could establish guidelines for its implementation, ensuring interoperability and scalability across industries. This would not only protect consumers from identity theft and unauthorized access but also bolster the integrity of digital ecosystems reliant on voice authentication.

AI/Cloud Regulatory “Safe Spaces” (Sandboxes 2.0) in Financial Services sector

Financial institutions increasingly would like to use AI for fraud detection and prevention, enhanced customer service, and to reduce costly back-office functions. To foster innovation in financial services while ensuring regulatory oversight, Pindrop recommends:

- The White House and relevant agencies in the financial sector should consider the establishment of innovative, federal regulatory "safe spaces" for testing advanced AI and cloud technologies. These controlled environments would allow financial institutions to benefit from AI innovation by experimenting with cutting-edge solutions, even those not easily suited to traditional on-premise and/or government-cloud-only sandboxes, without compromising consumer protection or market stability.

- By creating these “safe spaces”, regulators can partner with financial institutions to navigate compliance complexities, monitor systemic risks, and gather insights into emerging technologies.
- The benefits of this would be manifold. Financial institutions could leverage AI to enhance customer experiences, improve risk management, and streamline operations. These “safe spaces” would also encourage collaboration between regulators and industry leaders and enable the former to gain a greater understanding about how AI solutions actually operate.
- Ultimately, this approach would accelerate the adoption of transformative technologies and help ensure American leadership in two crucial sectors: financial services and AI.
- Pindrop applauds the Chairman of the House Financial Services Committee, Rep. French Hill, for having authored legislation to allow financial institutions the flexibility “to safely and soundly experiment with AI and enhance performance on behalf of our nation’s citizens.”²³

White House Authentication / Deepfake Summit

While federal leadership has largely helped mitigate the perils and abusiveness of email spam, problems with audio and video deception persist and grow. Therefore, to combat deepfakes and unleash and protect American innovation, Pindrop recommends:

- Convening an Authentication / Deepfake Summit that would bring together key stakeholders in the telecommunications and other key sectors to address the growing challenges of voice fraud and abusive deepfake audio attacks. This summit would focus on encouraging the development and widespread adoption of audio authentication and liveness detection technologies – including on the part of the U.S. Government - to enhance security across various mediums.
 - The summit would convene financial institutions, telecom providers, handset manufacturers, and deepfake detection authentication providers to collaborate on creating a more secure voice ecosystem. Participants would explore cutting-edge technologies such as liveness detection, which can analyze unique voice patterns and spectral distortions to distinguish between human and synthetic voices with up to 99% accuracy. The summit should review upgrading security protocols of voice and video interactions (telephony

²³ <https://hill.house.gov/news/documentsingle.aspx?DocumentID=9405>

communications, video conferencing, *etc.*) to mitigate deepfake risks against nation-state actors.

- o The goal would be to promote collaboration and innovation in this field and to drive toward a unified standard that can protect consumers and companies from sophisticated voice-based fraud attempts while maintaining seamless user experiences across various platforms and devices. Additional goals would be to enhance U.S. AI and deepfake detection dominance in order to promote economic competitiveness and national security.²⁴

Implementation of the Take It Down Act

Pindrop appreciates the leadership of the Administration in calling for the passage of the “Take it Down Act” during the State of the Union. This legislation criminalizes the publication of Non-Consensual Intimate Imagery (“NCII”) and mandates that social media and other platforms remove such content within 48 hours of notification from a victim.

- For this legislation to be effective, there must be a means of enforcement and factually determining a deepfake. Should the Take It Down Act be signed into law, Pindrop encourages the incentivization of compliance through – in part – the adoption of liveness detection solutions.
- Even though the bill penalizes only a certain subset of deepfakes - and governs the duties of platforms to take down content regardless of its veracity - it represents an important opportunity for the growth of liveness detection solutions that aid in identifying deepfakes.

²⁴China's handset manufacturer Honor has unveiled device with deepfake detection as well as announced they will invest \$10B in AI devices (see https://www.yahoo.com/tech/finally-theres-smartphone-built-detect-144911342.html?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlMmNvbS8&guce_referrer_sig=AQAAAKdnmyx0Me8Q-9XInyDLorLiVHQM1N8iL24WnbTcDDChiJcahNIZRwf3d7_O5ewgTKVneFABc6ZjWNHzrb9peJVqbtdPt3J8j-oxWLkqdsHTaanAa8SAEJZRvGnXogNQz_QFchS5svz5Jjm3Ru_gwclzZvUher7dkzgnwrTLlhqd; https://www.reuters.com/technology/artificial-intelligence/chinas-honor-announces-10-billion-investment-ai-devices-2025-03-02/). In the U.S., deepfake detection services struggle to protect citizens because access to this audio is not open.

4. Conclusion

As AI-generated scam calls and deepfakes become more prevalent, anti-fraud, authentication, and deepfake detection technologies will become ever more important. Pindrop welcomes the Administration's focus on promoting innovation and looks forward to collaborating with federal partners to deploy early deepfake detection capabilities to protect U.S. consumers and businesses.

Respectfully,

Clarissa Cerda
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Pindrop Security, Inc.

Rahul Sood
Chief Product Officer
Pindrop Security, Inc.